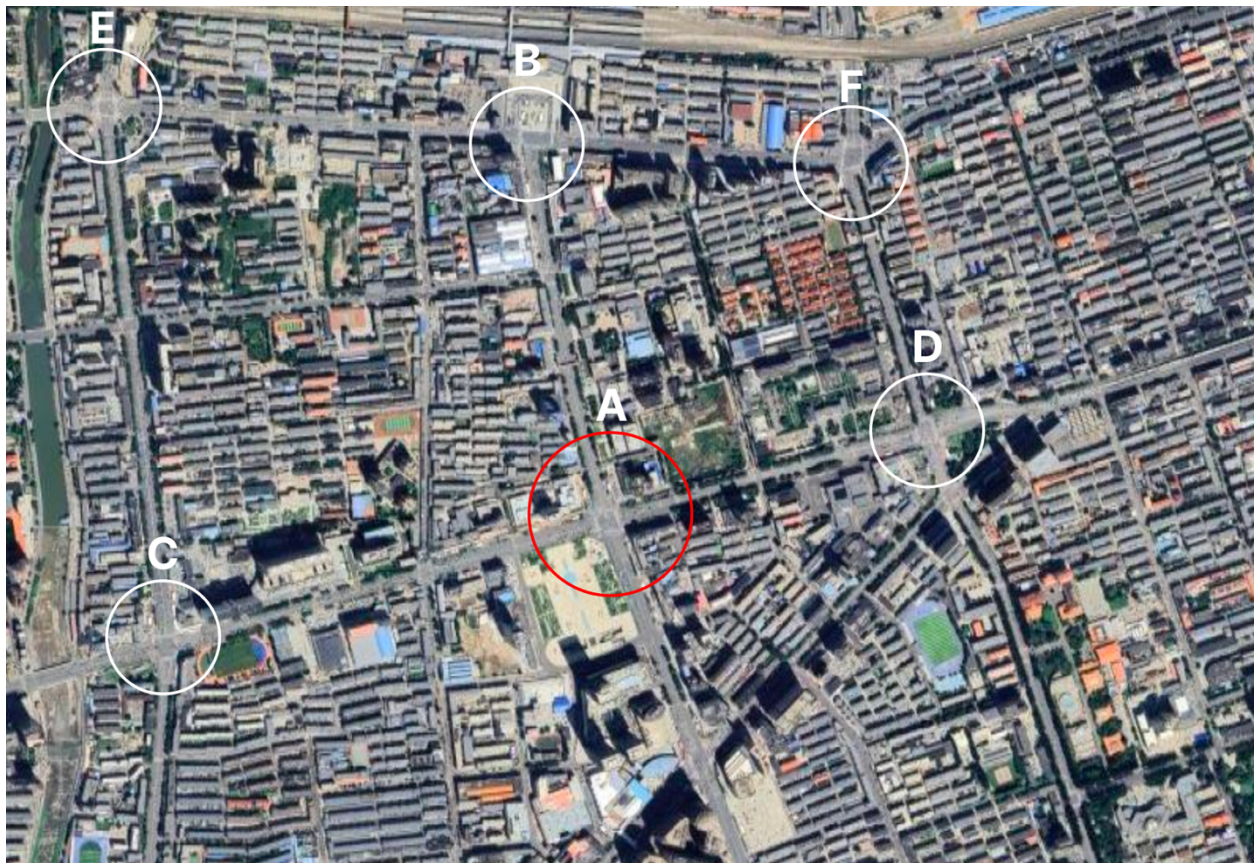


Road network selection

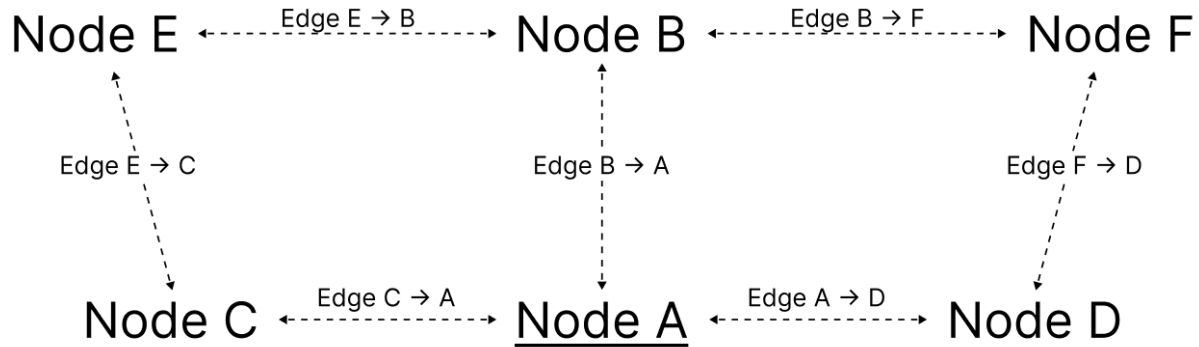
For this task, we selected the main junction marked as A in the satellite image. This is the central intersection that we focus on in our traffic analysis. However, traffic behavior is not limited only to this one junction. Nearby intersections can influence it because vehicles move between connected roads, and congestion can spread from one junction to another.

To represent this local traffic system, we also selected five surrounding intersections: B, C, D, E, and F. These intersections are located around the main junction and form a small local road network. This allows us to analyze traffic not as a single isolated point, but as a connected system.



Graph representation

The road network can be represented as a graph.



In this graph:

Nodes = intersections

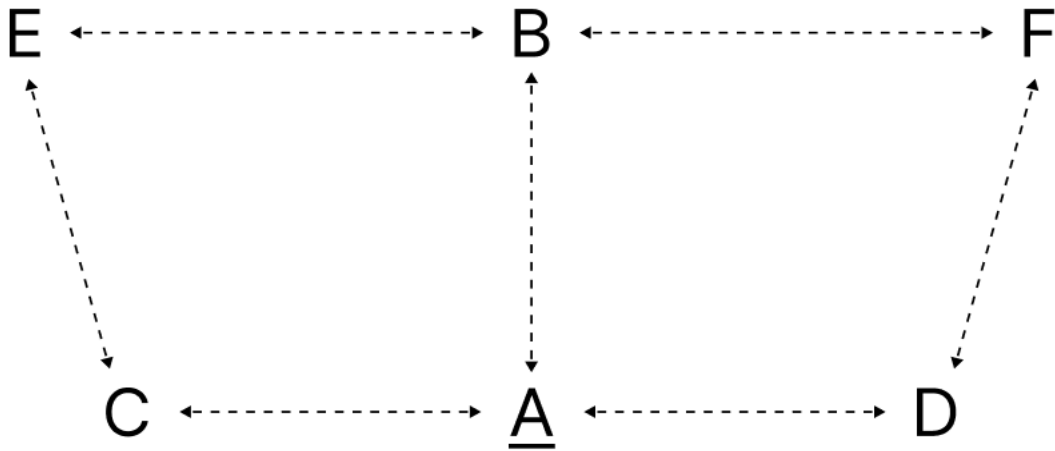
Edges = roads connecting intersections

The selected graph contains the following nodes:

Node	Description
A	Main selected junction
B	Northern nearby intersection
C	South-western nearby intersection
D	Eastern nearby intersection
E	North-western nearby intersection
F	North-eastern nearby intersection

The main idea is that each intersection becomes a node in the graph, and each road connection between two intersections becomes an edge.

A simplified graph representation of the area is:



This is a simplified version of the real road network. It does not need to show every small street. The goal is to represent the most important nearby intersections and the main road connections between them.

Edge representation

The edges represent the road segments that connect the selected intersections. For this local network, possible edges are:

Edge	Description
A-B	Road connection between the main junction and the northern intersection
A-C	Road connection between the main junction and the south-western intersection
A-D	Road connection between the main junction and the eastern intersection
B-E	Road connection between the northern and north-western intersections
B-F	Road connection between the northern and north-eastern intersections
C-E	Road connection between the south-western and north-western part of the area
D-F	Road connection between the eastern and north-eastern intersections

These edges show how traffic can move through the local network. For example, congestion at node B could influence node A because they are connected by a road. Similarly, congestion at node D or F could later affect traffic flow around the main junction.

Node features

Each node can contain information about the intersection itself. These are called node features.

For this project, possible node features are:

Node feature	Explanation
Vehicle count	Number of vehicles detected near the intersection
Junction type	Type of intersection, for example crossroad or T-junction
Traffic light presence	Whether the intersection has traffic lights
Number of incoming roads	How many roads enter the intersection
Number of outgoing roads	How many roads leave the intersection
Congestion level	Estimated traffic density at the intersection
Average waiting time	Estimated delay for vehicles at the junction
Detected vehicle categories	Number of cars, buses, trucks, motorcycles or bicycles

For example, node A is the main junction and would probably have the most important traffic information. It could include the number of detected cars, traffic density, queue length, and traffic light state.

Edge features

Each edge can contain information about the road segment between two intersections. These are called edge features.

Possible edge features are:

Edge feature	Explanation
Road length	Distance between two connected intersections
Number of lanes	Road capacity between the intersections
Speed limit	Maximum allowed speed on the road segment
Road direction	One-way or two-way traffic
Estimated traffic flow	Number of vehicles moving through the road segment
Travel time	Estimated time needed to move from one node to another
Road type	Main road, residential road, side road, etc.
Congestion risk	How likely congestion is to spread through this road

For example, the edge A-B represents the road between the main junction and the northern intersection. If this road has many vehicles and limited lanes, it may have a higher congestion risk.

How a GNN could be used

A Graph Neural Network, or GNN, could be used to model traffic behaviour in this road network.

A normal model usually looks at one intersection separately. However, traffic is connected. If one nearby intersection becomes congested, it can affect other intersections. A GNN is useful because it can learn from the connections between intersections.

In this traffic graph, each node starts with its own information, such as vehicle count, congestion level, traffic light presence, and junction type. Each edge contains road information, such as road length, number of lanes, speed limit, and traffic flow.

The GNN then passes information between connected nodes. This process is called message passing.

For example:

```
Node B sends traffic information to node A.  
Node C sends traffic information to node A.  
Node D sends traffic information to node A.  
Node F sends traffic information to node B and D.
```

This means that node A does not only use its own traffic information. It also receives information from nearby intersections. Because of this, the model can understand how the surrounding network affects the main junction.

After one GNN layer, each node knows information from its direct neighbors. After multiple layers, a node can also receive information from intersections that are further away. This is useful because congestion can spread through several connected roads.

What the GNN could predict

A GNN-based traffic model could predict several useful things:

Prediction	Explanation
Future congestion level	Predict whether an intersection will become busy in the next few minutes
Traffic flow	Estimate how many vehicles will pass through each road segment
Queue length	Predict how long the vehicle queue may become at traffic lights
Travel time	Estimate how long it will take to move between two intersections
Congestion propagation	Predict whether congestion at one junction will spread to another
Traffic light optimization	Suggest which direction should receive a longer green light phase

For example, if the GNN detects that nodes B and D are becoming congested, it may predict that node A will also become congested soon. This could help a smart traffic system adjust traffic lights before the congestion becomes worse.

Final explanation

The selected road network around junction A can therefore be represented as a graph with six nodes: A, B, C, D, E, and F. Each node represents an intersection, and each edge represents a road connection between intersections.

This graph structure is useful because traffic behavior depends on the surrounding network. A traffic jam at one intersection can influence nearby intersections through connected roads. A GNN could use node features and edge features to learn how traffic moves through the area and how congestion spreads over time.

Such a model could be used to predict future congestion, traffic flow, queue length, travel time, and possible congestion propagation. It could also support intelligent traffic light control by helping the system decide which roads need more green light time.